To: Office of the Deputy Secretary and ASTP/ONC, Department of Health and Human Services
From: Stephen Gabriel
Date: February 23, 2026
Re: Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care (Docket No. HHS-ONC-2026-0001)

In response to HHS's request for information on accelerating AI adoption in clinical care, I write to highlight structural barriers often invisible from hospital-centered policy perspectives. Dental care serves as a useful case study because it combines significant population health need, extreme software fragmentation, and systematic exclusion from federal health IT modernization efforts. The patterns identified here likely extend to other ambulatory settings including behavioral health, home health, and long-term care.

The current AI adoption problem in ambulatory settings traces directly to federal health IT policy decisions enacted nearly two decades ago. The HITECH Act of 2009 authorized approximately $27 billion to incentivize electronic health record adoption, operationalized through the Meaningful Use program. This program successfully drove hospital EHR adoption to near-universal levels and forced medical software vendors to adopt standardized data ontologies and exchange protocols.[1] However, the statutory framework created a wide gap between standard medicine and specialized ambulatory care. Dentists were explicitly excluded from the Medicare EHR Incentive Program. While theoretically eligible for the parallel Medicaid program, participation required demonstrating 30% Medicaid patient volume, a threshold most private dental practices could not meet because adult dental benefits are frequently carved out of state Medicaid programs.[2] The dental sector consequently received virtually no federal capital to subsidize the transition to digital records and no federal mandate to adopt certified, interoperable software.

Without this mandate, the dental software market evolved through localized free-market dynamics, resulting in severe fragmentation. The current landscape features major vendors like Dentrix, Eaglesoft, and Open Dental alongside a long tail of over 150 disparate clinical, imaging, and practice management systems operating without standardized data exchange protocols.[3] This fragmentation creates a significant barrier for AI developers. In the broader medical sector, an AI developer can use standard FHIR APIs to interface with dominant electronic health records, immediately scaling products to thousands of hospitals. A dental AI developer must build, test, and continuously maintain custom integration bridges for dozens of proprietary systems, many of which actively restrict third-party access or use proprietary data storage formats. This technical debt restricts scalability, inflates development costs, and leaves developers without the diverse, multi-site training data required to build robust algorithms.

The fragmentation of source data leads to a secondary problem in AI evaluation and post-market safety. The FDA's AI/ML-Enabled Medical Devices database shows over 1,300 authorized devices by end of 2025, with 295 new clearances in 2025 alone.[4] Approximately 52 of these are dental AI devices, predominantly for oral radiology applications including caries detection and cephalometric analysis.[5] However, most clearances occur via the 510(k) pathway, which does not require the extensive prospective clinical trials of Premarket Approval. Many 510(k) summaries lack detail on demographic representation of training data and scope of cross-site validation. A systematic review of federated learning in healthcare found that only 5.2% of studies reached real-world clinical implementation. This 94.8% failure rate was driven not by mathematical flaws but by organizational, infrastructural, and economic barriers.[6] Federated learning allows algorithms to train on distributed data without centralizing protected health information, which theoretically solves the privacy barrier. But it still requires computational overhead at edge nodes and baseline data harmonization that fragmented ambulatory systems cannot provide.

The FDA has recognized these lifecycle management challenges. Title 21 CFR Part 820, governing the Quality System Regulation for medical devices, underwent a major shift effective February 2, 2026, transitioning to the Quality Management System Regulation (QMSR) harmonized with ISO 13485:2016.[7] This framework imposes stricter documentation requirements for software validation and continuous post-market monitoring. For AI developers, compliance requires real-world performance tracking to detect algorithmic drift. Yet without standardized interoperability infrastructure in ambulatory settings, conducting effective post-market surveillance on dental AI tools is not currently feasible. The regulatory expectation for continuous monitoring does not match the reality of fragmented IT infrastructure.

Even if technical barriers could be overcome, the economic mechanism for AI adoption remains broken. The CDT code system maintained by the American Dental Association contains no codes for AI-assisted diagnostics.[8] The CDT 2026 update introduced codes for point-of-care saliva testing and cracked tooth diagnostics, but nothing for algorithmic image interpretation or predictive software analysis. Without billing codes, practices cannot submit claims to third-party payers for AI-enhanced services. The cost must be absorbed as overhead. This contrasts with the CPT system, where Category III codes now allow tracking of AI use in radiology and pathology, providing the foundation for eventual Medicare coverage determinations under 42 CFR Part 414.[9] The absence of dental AI codes creates a reimbursement problem that pushes private investment toward medical AI, where scalable payment pathways exist.

These structural barriers have direct public health consequences. HRSA data for Q1 FY2026 identifies 7,443 designated dental Health Professional Shortage Areas encompassing 63.7 million Americans, with only 32.93% of clinical need currently met.[10] AI-assisted diagnostic tools have exactly the characteristics required to address this crisis. A dental hygienist could capture radiographs and run them through an FDA-cleared triage system, flagging acute pathologies for prioritized intervention by the covering dentist. But the safety-net practices operating in these shortage areas are the very organizations most affected by the barriers identified here: least able to afford AI software without reimbursement, least likely to operate interoperable systems, and least represented in validation datasets that skew toward affluent urban academic centers.

Several targeted interventions would address these barriers. First, ONC should expand certification criteria under 45 CFR 170.315 to address dental workflows and imaging AI, potentially requiring certified software for clinics participating in Medicaid dental networks.[11] Second, the ADA CDT Committee should develop emerging technology codes for AI-assisted diagnostics, drawing precedent from CPT Category III codes, while CMS issues guidance to state Medicaid directors on coverage pathways. Third, FDA guidance on AI device lifecycle management under the new QMSR framework should include dental-specific examples given the unique variability of dental imaging acquisition. Fourth, ASTP/ONC should fund dental FHIR implementation guide development through HL7 cooperative agreements, integrating periodontal charting and dental imaging metadata into the US Core Data for Interoperability standard. Finally, FDA should sponsor federated benchmarking registries that evaluate algorithms against diverse datasets, including data from HPSA safety-net clinics, satisfying QMSR validation requirements while ensuring algorithms perform safely across underserved populations.

The explicit policy exclusion of specialized ambulatory settings from historical health IT initiatives has resulted in fragmented software ecosystems, poor interoperability, and nonexistent reimbursement pathways. These failures ensure that the diagnostic potential of AI remains concentrated in well-resourced academic medical centers, bypassing the populations who need it most. Unlocking clinical AI across the full healthcare continuum requires more than software innovation. It demands realignment of ONC certification standards, FDA lifecycle management protocols, and CMS payment mechanics.

Sincerely,

Stephen Gabriel

The views expressed above are solely those of the author and should not be attributed to any organization.

References

1. CMS, Medicare and Medicaid EHR Incentive Programs. The Meaningful Use program transitioned to Promoting Interoperability under MIPS.

2. HITECH Act of 2009, Medicaid EHR Incentive Program eligibility requirements. 30% Medicaid patient volume threshold over 90-day period.

3. Dental software market fragmentation documented in industry analyses. Major vendors lack standardized data exchange comparable to hospital EHR interoperability.

4. FDA Artificial Intelligence and Machine Learning (AI/ML)-Enabled Medical Devices database. https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-aiml-enabled-medical-devices

5. FDA 510(k) Premarket Notification Database, dental AI device clearances through early 2026.

6. Teo ZL, et al. Federated machine learning in healthcare: A systematic review on clinical applications and technical architecture. Cell Reports Medicine. 2024.

7. 21 CFR Part 820, Quality Management System Regulation (QMSR), effective February 2, 2026, harmonizing with ISO 13485:2016.

8. American Dental Association, CDT 2026 code set. 60 updates including 31 new codes; no codes for AI-assisted diagnostics.

9. 42 CFR Part 414 (physician fee schedule); 42 CFR Part 410 (supplementary medical insurance benefits).

10. HRSA Data Warehouse, Designated Health Professional Shortage Areas Quarterly Summary, Q1 FY2026. https://data.hrsa.gov/

11. 45 CFR 170.315, ONC Health IT Certification Criteria.